

LESSON 8-4 • SESSION 1 • PRACTICE • FORM A

Writing Inequalities

Learning target – I can write an inequality that describes a constraint in a real situation. • 6.EE.B.8

REMEMBER

- **At least** = that number or more ($\geq$). **More than** = greater, not equal ($>$).
- **At most, no more than, maximum** = that number or less ($\leq$). **Less than, fewer than** = $<$.
- Ask: is the boundary number **allowed**? Then choose the symbol.

WORKED EXAMPLE

A van can carry at most 8 riders. Write an inequality for r , the riders.

Step 1 Key words: “at most.” Is 8 allowed? Yes.

Step 2 “At most” means less than or equal to: $r \leq 8$.

Step 3 Test: 8 works, 5 works, 9 does not.

Answer $r \leq 8$

WORDS inequality (*desigualdad*) • at least (*al menos*) • constraint (*restricción*)

PRACTICE

show your work • label units

1. Write an inequality for each phrase. Use x .

a. at least 18 _____

b. at most 7 _____

c. more than 25 _____

d. fewer than 12 _____

e. no more than 15 _____

f. a minimum of 4 _____

2. Write an inequality for each rule. Use the variable given.

Rule	Inequality
A person must be at least 13 to see the film. (a)	
A project needs more than 85 points to earn a ribbon. (p)	
A lifeguard must train at least 30 hours. (h)	
A backpack may weigh no more than 15 pounds. (w)	

3. **Ticket Sales Goal.** A club meets its goal when it sells at least 150 tickets. Write an inequality for t . Name two solutions.

Hint “At least” lets 150 count.

Answer _____

4. **Phone Storage.** A game needs more than 9 GB of free space. Write an inequality for g . Does exactly 9 GB work?

Hint Is 9 itself allowed?

Show your work.

Answer _____

5. **Trip Chaperones.** Every bus needs at least 4 chaperones. One bus has 3. Write an inequality for c . Does this bus meet the rule?

Show your work.

Answer _____

- 6. Uniform Order.** A team gets a bulk price when it orders at least 25 shirts. Write an inequality for s . Name two solutions.

Hint *Decide whether 25 counts before you pick a symbol.*

Answer _____

- 7. Read It Back.** An elevator sign reads "Maximum 12 passengers."

Hint *"Maximum" means at most.*

a. Write an inequality for p , the passengers. _____

b. Are 12 passengers allowed? Explain. _____

- 8. Ride Rules.** Riders must be at least 42 inches tall and no taller than 76 inches. Let h = height in inches.

a. Write an inequality for the lower limit. _____

b. Write an inequality for the upper limit. _____

c. Name two heights that fit both rules and one that does not. _____

FIND THE ERROR

explain the mistake, then fix it

- 9. Malik's Work.** A sign reads: "You must be at least 48 inches tall to ride." Malik wrote $h > 48$. Explain his mistake. Write the correct inequality.

TEST PRACTICE

choose the best answer

- 10.** "At least 20" is written –

A $x > 20$

B $x \geq 20$

C $x < 20$

D $x \leq 20$

- 11.** "At most 9" is written –

A $x < 9$

B $x \leq 9$

C $x > 9$

D $x \geq 9$

SELF-CHECK CODE

find your answer · copy its letter into that problem's box · spell a math word

$c \geq 4$ M	$t \geq 150$ S	$h \geq 42$ L	$t > 150$ Q	$g > 9$ Y	$c > 4$ J	$g \geq 9$ Z	$s \geq 25$ B	$p \leq 12$ O	$p < 12$ X
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#3	#4	#5	#6	#7a	#8a
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CHECK YOURSELF

☐ I can do this on my own ☐ I need a little help ☐ I have a question about #____